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Period Doubling and Chaos in Unsymmetric Structures Under Parametric Excitation

Nonlinear oscillations of a single-degree-of-freedom, parametrically-excited system are considered. The stiffness involves quadratic and cubic nonlinearities and models a shallow arch or buckled mechanism. The excitation frequency is assumed to be close to twice the natural frequency of the system. Numerical integration is used to obtain phase plane portraits, power spectra, and Poincaré maps for large-time motions. Period-doubling bifurcations and several types of limit cycles and chaotic behavior are observed. Approximate analytical techniques are applied to analyze some of the limit cycles and transitions of behavior. The results are used to estimate the parameter region in which chaos may occur.

Introduction

Most studies of chaos in mechanical or structural systems have considered external (forcing) excitation (e.g., see Holmes and Moon, 1983). A typical example of the governing equation is Duffing's equation with damping and a harmonic forcing term on the right-hand side. In this paper, however, the excitation appears as a coefficient in the equation of motion (parametric excitation).

A number of earlier studies on chaos in parametrically-excited systems have treated a pendulum subjected to harmonic vertical excitation at its support (e.g., McLaughlin, 1981; Leven and Koch, 1981; Arneodo et al., 1983; Eckmann, 1983; Koch et al., 1983; Koch and Leven, 1985; Leven et al., 1985; Pompe and Leven, 1986). The governing equation in most of those works has the form

$$\ddot{\psi}(t)+k_1\dot{\psi}(t)+(k_2+k_3\cos\Omega t)\sin\psi(t)=0. \quad (1)$$

For example, the first of these references uses $k_1=0$ or 0.2, $k_2=1$, and $\Omega=2$; the second uses $k_1=0.15$, $k_2=1$, and $\Omega=1.56$; and the third uses $k_1=0.18$, $k_2=2$, and $\Omega=2$. Similar equations were analyzed by Gell and Nakach (1986) and Gitterman (1987).

The equation to be examined here is

$$\ddot{u}(t)+2c\dot{u}(t)+(\omega_0{}^2+P\cos\Omega t)u(t)+\alpha u^2(t)+\beta u^3(t)=0 \quad (2)$$

with parameter values

$$c=0.1, \quad \omega_0=1, \quad \alpha=3/2, \quad \beta=1/2 \quad (3)$$

and initial conditions

$$u(0)=0.02, \quad \dot{u}(0)=0. \quad (4)$$

In equation (2), c is the damping coefficient, ω_0 is the natural frequency of the undamped linearized system, P is the excitation amplitude, and Ω is the excitation frequency. This equation is unsymmetric in $u(t)$, due to the quadratic term. It is the equation of motion for a particular shallow arch subjected to horizontal and vertical pulsating loads, and for a two-bar mechanism containing a nonlinear spring (see the Appendix). Equation (2) with different sets of parameters was previously considered by Zavodney et al. (1989).

In the present paper, the parametric frequency Ω is assumed to be in the neighborhood of $2\omega_0$ (principal parametric resonance). Attention is focused on the asymptotic (large-time) behavior of $u(t)$. First, large-time responses are investigated using numerical integration. Examples of phase plane portraits, power spectra, and Poincaré maps are presented. In the (Ω, P) plane, regions associated with different types of resonance are delineated.

Next, some of the transition curves and limit cycles are examined by a perturbation technique and the method of harmonic balance. In particular, period doubling is considered, since it often precedes chaos. In the spirit of work on forced systems by Rudowski and Szemplińska-Stupnicka (1987) and Szemplińska-Stupnicka (1988), simple analytical criteria are obtained which indicate regions in which chaos and other "strange" behavior may occur.

Large-Time Responses

When the parametric excitation is absent, (2) has fixed points (equilibrium states) at $\dot{u}=0$, $u=0$, -1, and -2, which will be called the right, central, and left fixed points, respec-

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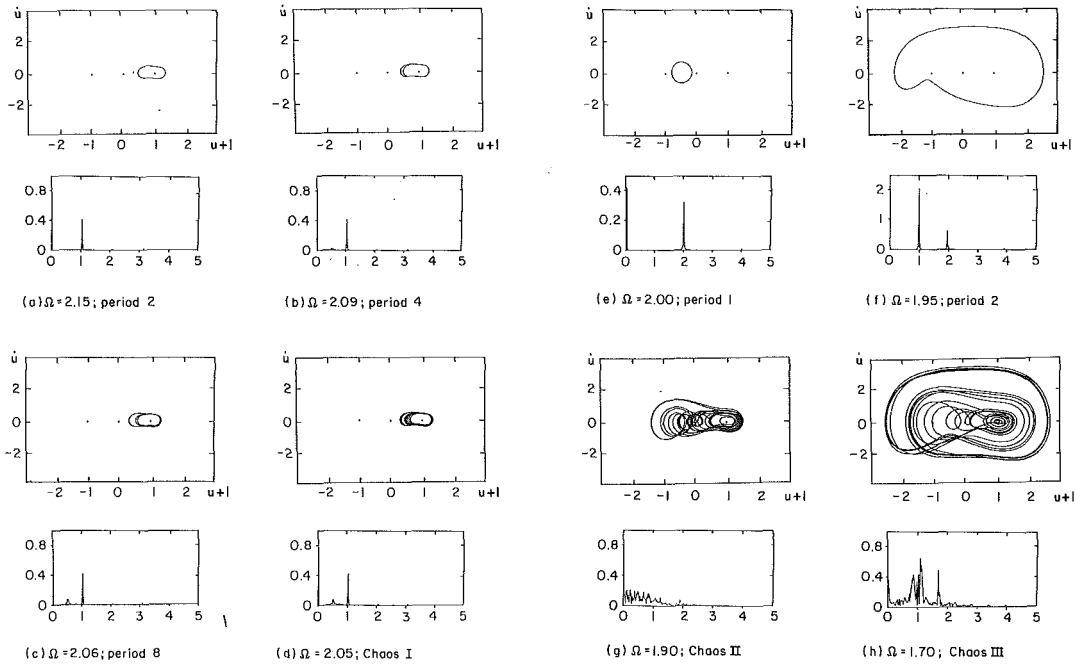


Fig. 1 Phase plane portraits and power spectra for $P = 0.9$ and various values of Ω; $x = u + 1$

tively. The outer ones are stable and the central one is unstable. Defining $x = u + 1$, so $x = 0$ at the central fixed point, (2) with parameters (3) can be written in terms of x as

$$\ddot{x}(t) + 0.2\dot{x}(t) - \frac{1}{2}(1 - 2P\cos\Omega t)x(t) + \frac{1}{2}x^3(t) = P\cos\Omega t. \tag{5}$$

If the parametric excitation term is dropped from (5), the resulting equation has been studied extensively, in part with reference to forced motion of a buckled beam (e.g., see Holmes, 1979; Dowell and Pezeshki, 1986).

Large-time responses are obtained by integrating (5) with initial conditions $x(0) = 1.02$ and $\dot{x}(0) = 0$, which is equivalent to solving (2) with parameters (3) and initial conditions (4). (Starting at $u(0) = \dot{u}(0) = 0$ would simply yield $u(t) \equiv 0$ and, hence, $u(0)$ is given a small initial value.) Some typical phase plane portraits and power spectra are depicted in Fig. 1, where the excitation amplitude is fixed at $P = 0.9$ and eight values of the excitation frequency are considered.

In Fig. 1(*a*), $\Omega = 2.15$ and the response, after transient behavior, settles into a limit cycle encircling the right fixed point ($u = 0$). The frequency of this large-time response is $\Omega/2$, as seen in its power spectrum. It is called a "period-2" motion, since its period is twice that of the excitation. A limit cycle such as that in Fig. 1(*a*) will be called a "right small orbit."

In Fig. 1(*b*), $\Omega = 2.09$ and the limit cycle has two loops and period 4. This change is called a "period-doubling bifurcation." The period is doubled again in Fig. 1(*c*), where $\Omega = 2.06$. When $\Omega = 2.05$, in Fig. 1(*d*), the response is chaotic rather than periodic. The trajectory remains close to the right fixed point, and this type of chaos will be called "Chaos I" here. The corresponding Poincaré map is shown in Fig. 2(*a*), where points are plotted in the phase plane during the large-time response at times nT, $T = 2\pi/\Omega$, for integer values n in a certain range. It indicates that the response is not periodic.

In Fig. 1(*e*), with $\Omega = 2.00$, the phase plane portrait involves a limit cycle lying between the left and central fixed points. Its period is the same as that of the excitation. This unusual limit cycle will be studied in a later section. In Fig. 1(*f*), $\Omega = 1.95$ and there is a large period-2 limit cycle encircling all three

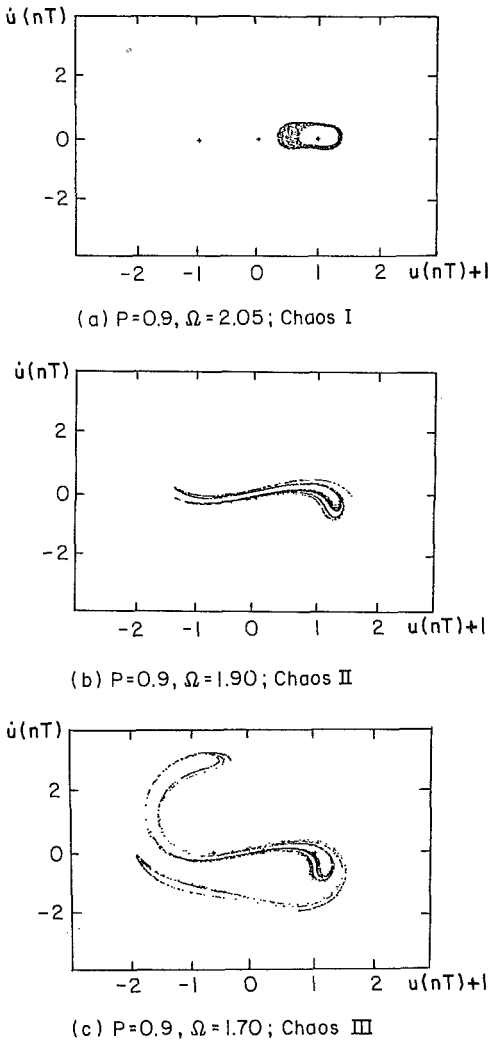


Fig. 2 Poincaré maps corresponding to Fig. 1(*d*), (*g*), and (*h*); $x = u + 1$

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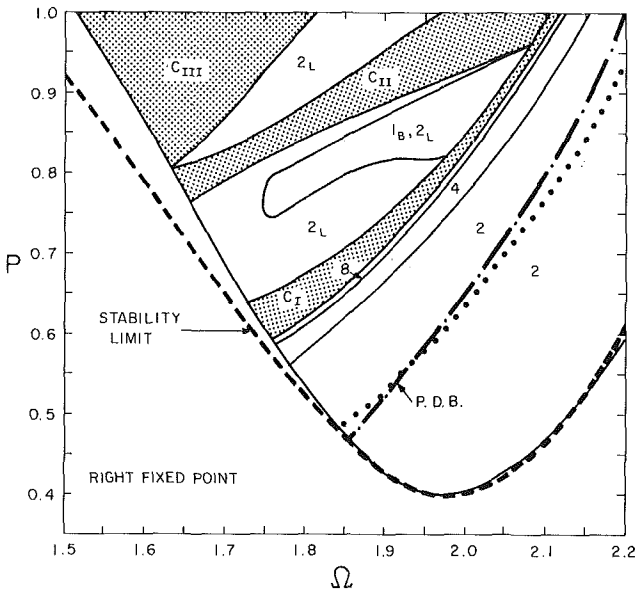


Fig. 3 Regions of different types of large-time response in (Ω, *P*) plane; P.D.B. denotes approximate conditions for period-doubling bifurcation based on (24) (dash-dot curve) and (25) (dotted curve)

fixed points. Chaotic behavior is seen again in Fig. 1(*g*) and Fig. 1(*h*), where Ω = 1.90 and Ω = 1.70, respectively. The corresponding Poincaré maps are presented in Fig. 2(*b*) and Fig. 2(c). The type of chaos in Fig. 1(*g*), which extends beyond the outer fixed points but does not have very large amplitude (e.g., in comparison to the limit cycle in Fig. 1(*f*)), will be called "Chaos II." Large-amplitude chaotic motion, as in Fig. 1(*h*), will be called "Chaos III."

Large-time responses were determined for many combinations of Ω and *P* in the ranges $1.5 \le \Omega \le 2.2$ and $0 < P \le 1.0$. The results are presented in Fig. 3 (where $0.35 \le P \le 1.0$), with solid curves separating regions having different types of behavior. If *P* is sufficiently small, the motion approaches the right fixed point (i.e., $u(t) \to 0$). The boundary of this bottom region has a minimum near (Ω, *P*) = (1.96, 0.40). In the regions labeled 2, 4, and 8, the responses are similar to those illustrated in Fig. 1(*a*), (*b*), and (*c*), respectively. The symbols C_I, C_{II}, and C_{III} denote Chaos I, II, and III, respectively, as in Fig. 1(*d*), (*g*), and (*h*). Large motions of period 2, as in Fig. 1(*f*), are indicated by 2_L. Finally, the symbol 1_B refers to a limit cycle of period 1 which lies between the left and central fixed points, as in Fig. 1(*e*). In the region which contains both symbols 2_L and 1_B, these two types of limit cycles are interspersed and a simple boundary separating them was not obtained.

Some comparisons can be made between these results, based on (2), and the previous studies of the parametrically-forced pendulum, based on (1) with $k_1 \ge 0$ and $k_2 > 0$. The fixed points of (1) are at $\dot{\psi} = 0$, $\psi = \pm n\pi$, $n = 0, 1, 2, \ldots$, and "rotating motions" (ψ increasing monotonically) may exist. As k_3 was increased, period-doubling bifurcations occurred, followed by chaos. A local type of chaos was mentioned in Arneodo et al. (1983) and was discussed in Leven et al. (1985) where a Coulomb damping term was added to (1). The nonrotating limit cycles began with period 2 as k_3 was increased from zero, while the rotating motions first appeared with period 1. Small "windows" of periodic solutions were sometimes found in the chaotic regime (e.g., Leven et al., 1985). Due to the rotating motions the response of (1), as described in the references, is quite different from that of (2), although some characteristics (such as the period-doubling route to chaos) are similar.

It would be useful to be able to approximate some of the curves in Fig. 3 by analytical means. In particular, a "lower bound" on the regions of chaos would be of interest. This will be examined in the following sections. In addition, an analysis of the amplitudes of some of the types of limit cycles will be carried out.

Stability of Right Fixed Point

Inside the bottom region of Fig. 3, the right fixed point is asymptotically stable. In order to determine when it loses stability, small motions about it are considered. Writing $u(t) = \xi(t)$, where $\xi(t)$ is small, and linearizing (2), one obtains

$$\ddot{\xi}(t) + 2c\dot{\xi}(t) + (\omega_0^2 + P\cos\Omega t)\xi(t) = 0. \tag{6}$$

From Floquet theory, one expects $\xi(t)$ to have period 2 at the stability limit in the neighborhood of $\Omega = 2\omega_0$. A first approximate solution can be obtained by assuming

$$\xi(t) = e^{\mu t}[b_{1c}\cos(\Omega t/2) + b_{1s}\sin(\Omega t/2)] \tag{7}$$

with $\mu > 0$ in the region of instability. Putting $\mu = 0$ and applying the method of harmonic balance, one obtains the relation

$$\Delta \equiv (\Omega^2 - 4\omega_0^2)^2 + 16c^2\Omega^2 - 4P^2 = 0 \tag{8}$$

to be satisfied at the stability limit, with $\Delta < 0$ when $\mu > 0$ (Hayashi, 1985). Equation (8) is plotted as a dashed curve in Fig. 3 with $\omega_0 = 1$, $c = 0.1$. This approximate stability limit for the right fixed point can be used as an approximate boundary of the region in which the response decays and $u(t) \to 0$.

Right Small Orbit

In this section small period-2 limit cycles, such as the one in Fig. 1(*a*), will be examined. A perturbation technique will be applied to (2) in the neighborhood of $\Omega = 2\omega_0$ to obtain an approximate solution for $u(t)$.

Let

$$(\Omega/2)^2 = \omega_0^2 + \epsilon^2\sigma,\ c = \epsilon^2\tilde{c},\ P = \epsilon^2 R,\ \alpha = \epsilon\tilde{\alpha},\ \beta = \epsilon^2\tilde{\beta}, \tag{9}$$

where ϵ is a small parameter and σ is a detuning parameter. (Alternatively, one could let $u = \epsilon v$ and not define α and β.) Equation (2) becomes

$$\ddot{u} + (\Omega/2)^2 u + \epsilon\tilde{\alpha}u^2 + \epsilon^2(\tilde{\beta}u^3 + Ru\cos\Omega t + 2\tilde{c}\dot{u} - \sigma u) = 0. \tag{10}$$

Next, $u(t)$ is written in the form

$$u(t) = u_0(t) + \epsilon u_1(t) + \epsilon^2 u_2(t) + \ldots. \tag{11}$$

Expansion (11) is substituted into (10) and the coefficients of ϵ^0, ϵ^1, and ϵ^2 are set equal to zero. This yields the equations

$$\ddot{u}_0 + (\Omega/2)^2 u_0 = 0, \tag{12}$$

$$\ddot{u}_1 + (\Omega/2)^2 u_1 = -\tilde{\alpha}u_0^2, \tag{13}$$

$$\ddot{u}_2 + (\Omega/2)^2 u_2 = -2\tilde{\alpha}u_0 u_1 - \tilde{\beta}u_0^3 - Ru_0\cos\Omega t - 2\tilde{c}\dot{u}_0 + \sigma u_0. \tag{14}$$

The solution of (12) can be written as

$$u_0(t) = a\cos\theta(t), \quad \theta(t) = (\Omega/2)t + \phi. \tag{15}$$

Then (13) has the solution

$$u_1(t) = -\alpha a^2[3 - \cos 2\theta(t)]/6. \tag{16}$$

Equations (15) and (16) are substituted into (14), assuming *b* and ϕ are independent of *t*. Trigonometric identities are utilized. The coefficients of $\sin\theta$ and $\cos\theta$ on the right-hand side of the resulting equation are then set equal to zero, in order to avoid secular terms. After using the values of α and β from (3), one obtains

$$a^2 = (4 - \Omega^2 \pm 2\sqrt{P^2 - 4c^2\Omega^2})/6, \tag{17}$$

$$\sin 2\phi = 2c\Omega/P, \quad \cos 2\phi = (\Omega^2 + 6a^2 - 4)/(2P), \tag{18}$$

and, from (11), (15), and (16),

$$u(t) = a\cos\theta(t) - (3/4)\epsilon a^2 + (1/4)\epsilon a^2\cos 2\theta(t) + 0(\epsilon^2) \tag{19}$$

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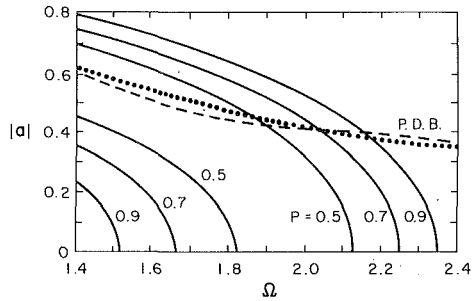


Fig. 4 Amplitude parameter in approximate period-2 response (19); P.D.B. denotes approximate conditions for period-doubling bifurcation based on (24) (dashed curve) and (25) (dotted curve)

where θ is defined in (15). Equation (19) furnishes an approximate solution for the period-2 motion associated with the right small orbit.

The absolute value $|a|$ of the response amplitude parameter in (19) is plotted as a function of the excitation frequency Ω in Fig. 4 for fixed excitation amplitudes $P = 0.5$, 0.7, and 0.9. When $\Omega = 2.15$, the upper branch for $P = 0.9$ yields $|a| = 0.40$, which agrees well with the amplitude 0.41 of the spike on the response spectrum in Fig. 1(*a*) at frequency $\Omega/2$.

The lower branches in Fig. 4 exhibit the classical (first-order) instability which is usually obtained by perturbation methods (Nayfeh and Mook, 1979, p. 347). The upper branch does not exhibit this type of instability. Other types of instability are considered in the following section.

Period-Doubling Bifurcations

In order to study higher-order instabilities, a variation $\eta(t)$ is added to the solution. Let

$$x(t) = 1 + u(t) + \eta(t) \tag{20}$$

where $u(t)$ is given by (19) and $\eta(t)$ is small. This form of $x(t)$ is substituted into (5) and the resulting equation is linearized in $\eta(t)$. One obtains the variational equation

$$\ddot{\eta}(t) + 0.2\dot{\eta}(t) + \left[\lambda_0 + \gamma_2 \sin 2\theta(t) + \sum_{j=1}^{4} \lambda_j \cos j\theta(t)\right]\eta(t) = 0 \tag{21}$$

where

$$\lambda_0 = 1 - (3/2)a^2 + (57/64)a^4, \quad \lambda_1 = 3a - (15/8)a^3,$$
$$\lambda_2 = (3/2)a^2 - (9/16)a^4 + P\cos 2\phi, \quad \lambda_3 = (3/8)a^3, \tag{22}$$
$$\lambda_4 = (3/64)a^4, \quad \gamma_2 = P\sin 2\phi,$$

$\theta(t)$ is defined in (15), and ϕ, a, and P satisfy (17) and (18). Equation (21) is a form of Hill's equation (Hayashi, 1985), and the period of the excitation in (21) is the same as the period of the solution $u(t)$ under investigation, i.e., $2T$ where $T = 2\pi/\Omega$.

To determine a higher-order instability, Floquet theory is applied. A solution of (21) is sought in the form

$$\eta(t) = e^{\mu t}\Phi(t) = e^{\mu t}\Phi(t + 4T)$$
$$= e^{\mu t} \sum_{k=1,3,5,\ldots}^{\infty} d_k \cos[(k\Omega t/4) + \delta_k]. \tag{23}$$

When $\mu > 0$, this type of solution leads to a growth of harmonic components with period $4T$ (i.e., twice the period of the solution u), and thus may lead to period-doubling bifurcation (Hayashi, 1985; Szemplińska-Stupnicka, 1987).

For an approximate solution, at the stability limit ($\mu = 0$), the first two terms of the expansion (23) are considered, written as

$$\eta(t) = d_{1c}\cos(\Omega t/4) + d_{1s}\sin(\Omega t/4) + d_{3c}\cos(3\Omega t/4) + d_{3s}\sin(3\Omega t/4). \tag{24}$$

Equation (24) is substituted into (21) and the method of harmonic balance is applied. This results in combinations of Ω, a, and P which are plotted as curves labeled P.D.B. in Fig. 3 (dash-dot) and Fig. 4 (dashed).

The dash-dot curve in Fig. 3 gives a rough approximation of the solid transition curve from period-2 to period-4 responses which was found from numerical integration of (2) with (3) and (4). Apparently more terms must be used in the expansion (23), leading to greater computational complexity, if a better agreement is desired. On the other hand, one can obtain a simple analytical approximation for the occurrence of period-doubling bifurcations by letting $d_{1s} = d_{3s} = 0$ in (24), which means the damping term in (21) is neglected. The method of harmonic balance then yields

$$\Omega^2 = 4(G \pm \sqrt{G^2 - J}) \tag{25}$$

where

$$G = -(20\lambda_0 + 17\lambda_1 + \lambda_3 + 8\lambda_5)/7$$
$$J = 4[(\lambda_1 + \lambda_5)^2 - (2\lambda_0 + \lambda_1)(2\lambda_0 + \lambda_3)]/7, \tag{26}$$
$$\lambda_5 = (9/2)a^2 - (9/16)a^4 - 2,$$

and λ_0, λ_1, and λ_3 are defined in (22). Equation (25), along with (17), leads to the dotted curves in Figs. 3 and 4. They are quite close to the curves based on the full expression (24).

Consider the V-shaped region in Fig. 3 lying above the dashed curve, obtained from (8), and the dash-dot or dotted curve. It is expected that chaos and other "strange" behavior will not occur below this region. Hence, the "V" formed by these curves furnishes a "lower bound" for chaotic behavior. This is similar to results for systems with external excitation presented by Rudowski and Szemplińska-Stupnicka (1987) and Szemplińska-Stupnicka (1988), although in those cases period-1 motions exist below the V-shaped region.

In Fig. 4, the response amplitude is associated with the approximate response (19). For sufficiently low values of $|a|$, the upper branches in Fig. 4 are stable. The dashed curve and the dotted curve give approximations to values of $|a|$ at which the period-2 motion becomes unstable and bifurcates into a period-4 motion.

Period-1 Limit Cycle

For one-degree-of-freedom systems such as (2), limit cycles usually encircle one or more fixed points in the phase plane. In Fig. 1(*e*), however, the limit cycle lies between the left and central fixed points. This behavior is particularly interesting and will be analyzed now.

The power spectrum in Fig. 1(*e*) exhibits a constant component and a component with the frequency of the excitation. Thus the corresponding motion $x(t)$ can be approximated by the expression

$$x(t) = A_0 + A_1\cos(\Omega t + \zeta) \tag{27}$$

where $-1 < x < 0$ for the responses under consideration. Equation (27) is substituted into (5) and the method of harmonic balance is applied. This results in the three equations

$$0.2\Omega A_1 + P(1 - A_0)\sin\zeta = 0, \tag{28}$$
$$(2A_0^2 + 3A_1^2 - 2)A_0 + 2PA_1\cos\zeta = 0, \tag{29}$$
$$(12A_0^2 + 3A_1^2 - 8\Omega^2 - 4)A_1 + 8P(A_0 - 1)\cos\zeta = 0. \tag{30}$$

For $P = 0.8$, 0.9, and 1.0, solutions of (28)–(30) for A_0 and A_1 are plotted in Fig. 5 as functions of Ω. The amplitude A_0 is negative, and the solid curves give $-A_0$.

For these values of P, one sees in Fig. 5 that solutions of (28)–(30) exist if Ω is larger than 1.76, 1.96, and 2.15, respectively. The first two of these (Ω, P) combinations lie on the up-

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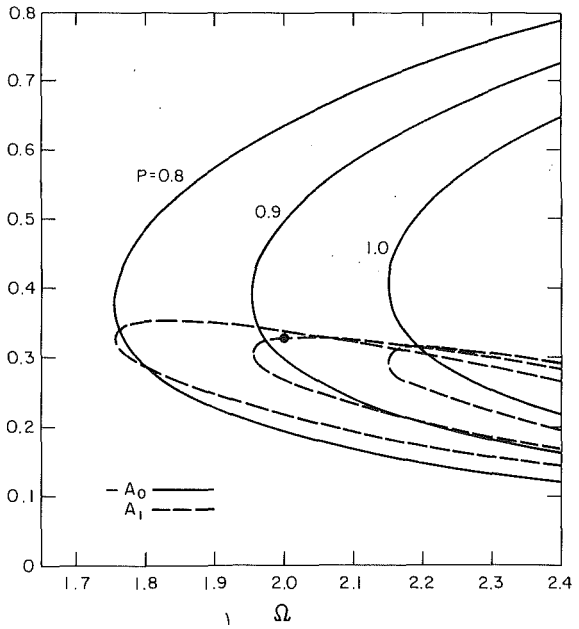


Fig. 5 Amplitudes in approximate period-1 response (27)

per boundary of the region denoted "1_B, 2_L" in Fig. 3. When this type of limit cycle was found in the direct numerical integration of (5), such as in Fig. 1(*e*), its response spectrum possessed amplitudes at frequencies zero and Ω which agree well with values on the upper branches of the $-A_0$ and A_1 curves in Fig. 5. One such value from Fig. 1(*e*), $A_1 = 0.33$ at $P = 0.9$ and $\Omega = 2.00$, is shown as a dot in Fig. 5. (Of course, the existence of such a stable limit cycle solution to (5) does not imply that the motion will approach it; that depends on whether or not the initial conditions are in the domain of attraction of that solution.)

Concluding Remarks

A one-degree-of-freedom system, governed by (2) and (3), has been considered. The equation of motion (2) contains a parametric excitation with amplitude P and frequency Ω which is assumed to be in the neighborhood of $2\omega_0$, where ω_0 is the natural frequency of the system. Linear viscous damping and quadratic and cubic nonlinearities in the stiffness are also included in (2). As described in the Appendix, the system can model a shallow arch or a mechanism subjected to special combinations of axial and transverse loads. These systems have two stable equilibrium states (fixed points) and an intermediate unstable one. Motions beginning near the right fixed point in the phase plane have been investigated here.

Depending on the excitation amplitude and frequency, various types of large-time responses are possible, as illustrated in Figs. 1-3. Three types of chaotic behavior have been described (see Fig. 1(*d*), (*g*), and (*h*), and Fig. 2). In type I, there is no "snap-through" to the left fixed point. In types II and III, the motion passes back and forth between the neighborhoods of the stable fixed points, with much larger amplitudes occurring in type III than in type II.

Approximate analytical methods have been utilized to examine some of the responses and their stability. The stability limit for the right fixed point gives an indication as to when the motions decay (dashed curve in Fig. 3). A theoretical lower bound on chaotic behavior in Fig. 3 is furnished by the V-shaped curve whose left side is the stability limit given by (8) and whose right side is the dash-dot curve (or, for a simple analytical result, the dotted curve determined from (17) and (25)). It is surprising that such a good lower bound can be obtained from such low-order approximations and from such simple analytical formulas as (8), (17), and (25).

Critical dynamic loads for curved or buckled structures are normally defined as the lowest loads at which snap-through occurs. For harmonic loading, the locus of these critical loads in the (Ω, P) plane (Plaut and Hsieh, 1985) tends to give another lower bound for chaos. However, chaos of type I may occur before the system exhibits motions past the central fixed point, and therefore this lower bound may not always be valid.

The type of limit cycle seen in Fig. 1(*e*) and denoted 1_B in Fig. 3 has been analyzed. It is the only period-1 large-time response seen in the numerical integration results in Fig. 3, and is unusual because it does not encircle any fixed point. Other period-1 responses would occur under different initial conditions. For example, if the excitation amplitude P is small and the system begins near the left fixed point, the large-time response is a small orbit encircling that point with period 1. This is understandable since the equation of motion about that point contains both parametric and external excitations, as in (5). Attention here, however, has been focused on the parametrically-excited system (2) with small initial conditions (4). Large motions of this system have not been studied extensively before, and it is seen that they exhibit interesting behavior which can be quite different from that of externally-excited systems.

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APPENDIX

First, consider the shallow elastic arch depicted in Fig. 6(*a*). The ends ($X = 0, L$) are simply supported, with the right end attached to a horizontal spring with stiffness coefficient k and allowed to move horizontally. The initial thrust is H_0 and the equilibrium configuration is $Y_0(X)$, while E denotes the modulus of elasticity, A is the cross-sectional area, I is the moment of inertia, r is the radius of gyration $\sqrt{I/A}$, m is the mass per unit length, and C is a damping coefficient.

Two dynamic loads are applied, as shown in Fig. 6(*b*): $\bar{Q}(X, \bar{t})$ is a distributed vertical load and $\bar{P}(\bar{t})$ is a horizontal load at the right support, with $\bar{t}$ denoting time. The configuration at time $\bar{t}$ is $Y(X, \bar{t})$. Define the nondimensional quantities

$$x = X/L,\ y = Y/(2r),\ y_0 = Y_0/(2r),\ t = (\bar{t}/L^2)\sqrt{EI/m},$$
$$h_0 = H_0L^2/(EI),\ p = \bar{P}L^2/(EI),\ q = \bar{Q}L^4/(2EIr),$$
$$c = CL^2/\sqrt{EIm},\ \gamma = k/[k + (EA/L)]. \tag{A1}$$

The equation of motion is given by

$$\frac{\partial^2 y}{\partial t^2} + 2c\frac{\partial y}{\partial t} + \frac{\partial^4 y}{\partial x^4} - \frac{d^4 y_0}{dx^4} + (p + h_0)\frac{\partial^2 y}{\partial x^2} - h_0\frac{d^2 y_0}{dx^2}$$
$$+ 2\gamma\frac{\partial^2 y}{\partial x^2}\int_0^1 \left[\left(\frac{dy_0}{dx}\right)^2 - \left(\frac{\partial y}{\partial x}\right)^2\right]dx = -q \tag{A2}$$

(Fung and Kaplan, 1952).

Assume that

$$y_0(x) = \lambda \sin \pi x,\ q(x,t) = \bar{q}(t)\sin \pi x \tag{A3}$$

and write the motion as

$$y(x,t) = [\lambda + u(t)]\sin \pi x. \tag{A4}$$

Substitution of (*A*3) and (*A*4) into (*A*2) leads to

$$\ddot{u} + 2c\dot{u} + \pi^2(\pi^2 + 2\pi^2\lambda^2\gamma - h_0 - p)u + 3\pi^4\gamma\lambda u^2 + \pi^4\gamma u^3$$
$$= \pi^2\lambda p - \bar{q} \tag{A5}$$

after dividing by sin πx. Equation (2) with parameters (3) is then obtained for the special case

$$h_0 = \pi^2,\quad \lambda = 1,\quad \gamma = 1/(2\pi^4),\quad c = 0.1,$$
$$p(t) = -(P/\pi^2)\cos\Omega t,\quad \bar{q}(t) = -P\cos\Omega t. \tag{A6}$$

A second example is provided by the mechanism in Fig. 7. It consists of two rigid bars with length L, a mass M at the central joint, a nonlinear (hardening) spring with linear stiffness coefficient K_1 and cubic coefficient K_3, and a dashpot with damping coefficient C. A horizontal load Q_0, larger than the buckling load $K_1L/2$, is applied, and then horizontal and vertical dynamic loads $\bar{Q}\cos\bar{\Omega}\bar{t}$ and $\bar{P}\cos\bar{\Omega}\bar{t}$ are applied as shown in Fig. 7(*b*). The vertical motion of the mass from the buckled

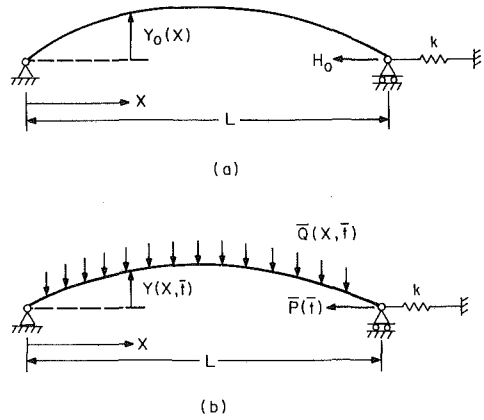


Fig. 6 Geometry of arch

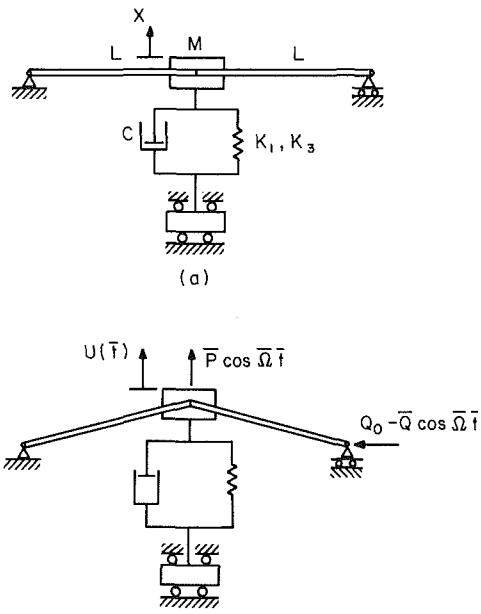


Fig. 7 Geometry of mechanism

equilibrium state is denoted $U(\bar{t})$, and it is assumed that the vertical deflection $X(\bar{t})$ from the unloaded horizontal equilibrium state is small enough so that $L^2 - X^2 \approx L^2$.

Define

$$D^2 = (2Q_0 - K_1L)/(K_3L),\ F^2 = ML/(4Q_0 - 2K_1L) \tag{A7}$$

and the nondimensional quantities

$$u = U/D,\ t = \bar{t}/F,\ c = CF/M,\ P = \bar{P}F^2/(MD),$$
$$Q = 2\bar{Q}F^2/(ML),\ \Omega = \bar{\Omega}F. \tag{A8}$$

If $Q = P$, the nondimensional equation of motion for $u(t)$ is given by (2) with parameters (3) (Plaut and Hsieh, 1987). (If Q and P were not equal, the equation of motion for $u(t)$ would include both parametric and external excitations.)

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